

Experiment 4- MHA (Mail Header Analyzer)

Done by:99240041247

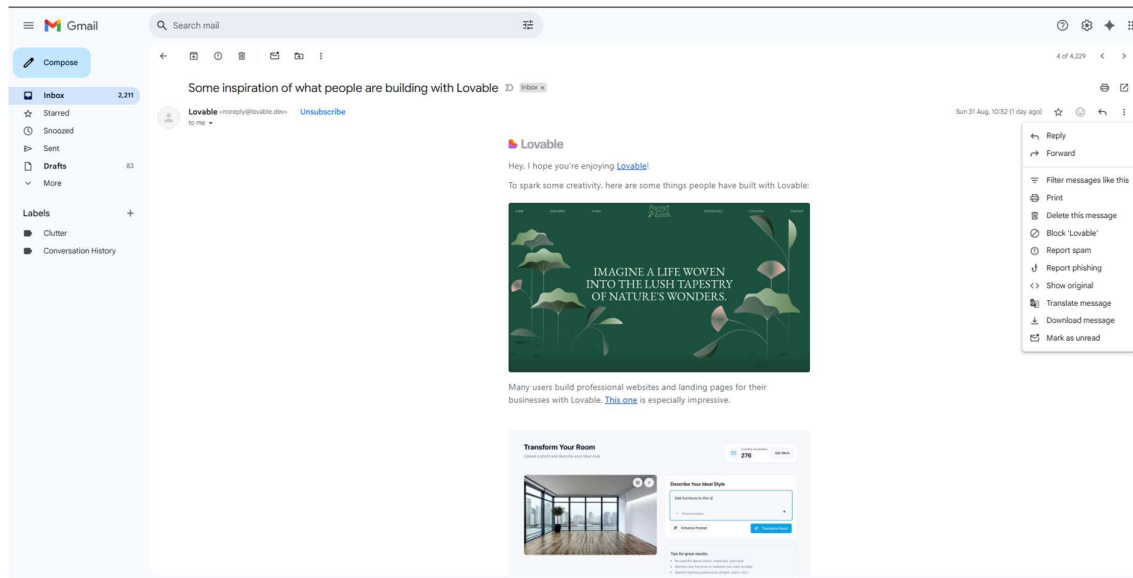
Aim :

- The aim is to use a Mail Header Analyzer (MHA) to trace an email's origin and verify its authenticity by examining its header for signs of spoofing.

Procedure

Step 1: Get the Email Header

- First, you need to copy the full, raw header from the email.
- Gmail: Open the email, click the three dots (:), and select Show original.

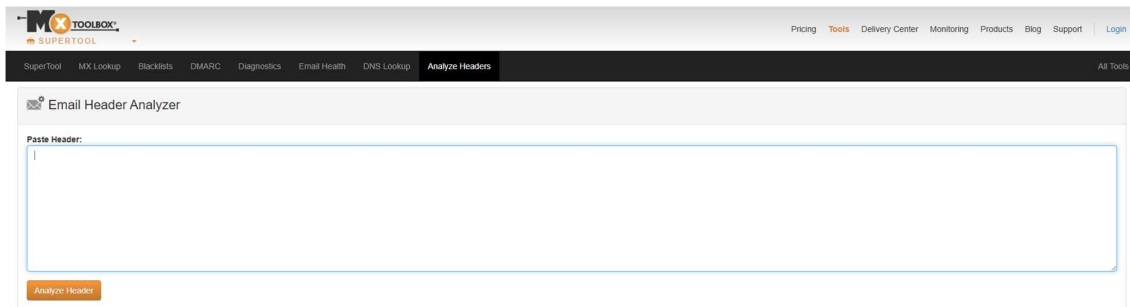


- Select all the text in the header and copy it.

```
Delivered-To: sudhama07pm@gmail.com
Received: by 2002:a05:7021:110:b0:b7:c16:c9dc with SMTP id 16csp511806dlk;
Sat, 30 Aug 2025 22:22:58 -0700 (PDT)
X-Google-Smtp-Source: AGH1iFNI1j1mxtV2F841GpKdZ5F28QvuaQmD5atbzy0E21F0r4e1OgR3YpfyRx/28qnJSK8m
X-Received: by 2002:a05:6000:2283:10:10:0:221:8922 with SMTP id fFacd0085a97d-3d1e04b0dc0e-2520631f8f.46.1756617778465;
Sat, 30 Aug 2025 22:22:58 -0700 (PDT)
ARC-Seal: i=1; a=rsa-sha256; t=1756617778; cv=none;
d=google.com; s=arc-20240605;
b=08U1U6SjYfSfSVVtq12Zou1J5RE8d94TutvrbjX64Dv72pTahVvH59h2xsmUCpEq
mzTHQ2xHovSm012chf60B6H937XGf8AP462qpsCVf2348Avhu72puXY7owRqIXbaF
wHwMU/235oofH7eazoeD7BQqFfz37D5P3313nkGdZTRSVhyhM6I05d1340Fbt58
WdQtw4w6m557WuVCln0e0575505p1pJQ/Cu4R0NIDK5e5NwA-QXcWpPha1hA6AK
WASAFIucbusw5qZ/kpxKRnmw/e1P06+2Uou22aJS05d6ZuXDC7J6jpoQhdt0Ulp4
Vv8g==
ARC-Message-Signature: i=1; a=rsa-sha256; c=relaxed/relaxed; d=google.com; s=arc-20240605;
h=message-id:1list-unsubscribe-post:1list-unsubscribe-to:1from:1subject:
1mime-version:1date:1dkim-signature:1dkim-signature;
bh=P5z3d038F5K16hLPj5dtr/NF1u0R2ZE3hdU5PQ2anU+;
b=08U1U6SjYfSfSVVtq12Zou1J5RE8d94TutvrbjX64Dv72pTahVvH59h2xsmUCpEq
mzTHQ2xHovSm012chf60B6H937XGf8AP462qpsCVf2348Avhu72puXY7owRqIXbaF
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WASAFIucbusw5qZ/kpxKRnmw/e1P06+2Uou22aJS05d6ZuXDC7J6jpoQhdt0Ulp4
Vv8g==
ARC-Authentication-Results: i=1; mx.google.com;
dkim=pass header.i=@clou0109333.lovable.dev header.s=mta header.b=MDG7LRk;
dkim=pass header.i=@u.mailgun.org header.s=kr2 header.b=FB0J40T;
spf=pass (google.com: domain of postmaster@clou0109333.lovable.dev designates 143.55.239.140 as permitted sender) smtp.mailfrom=postmaster@clou0109333.lovable.dev;
dmarc=pass (p=QUARANTINE sp=QUARANTINE dis=NONE) header.from=lovable.dev
Return-Path: <postmaster@clou0109333.lovable.dev>
Received: from a239-140.eu.mailgun.net (a239-140.eu.mailgun.net. [143.55.239.140])
by mx.google.com with UTF8SMTPS id fFacd0085a97d-3cf34067ba0b13811044f8f.696.2025.08.30.22.22.58
for <sudhama07pm@gmail.com>
(version=TLS1_2 cipher=ECDHE-ECDSA-AES128-GCM-SHA256 bits=128/128);
Sat, 30 Aug 2025 22:22:58 -0700 (PDT)
Received-SPF: pass (google.com: domain of postmaster@clou0109333.lovable.dev designates 143.55.239.140 as permitted sender) client-ip=143.55.239.140;
Authentication-Results: mx.google.com;
dkim=pass header.i=@clou0109333.lovable.dev header.s=mta header.b=MDG7LRk;
dkim=pass header.i=@u.mailgun.org header.s=kr2 header.b=FB0J40T;
spf=pass (google.com: domain of postmaster@clou0109333.lovable.dev designates 143.55.239.140 as permitted sender) smtp.mailfrom=postmaster@clou0109333.lovable.dev;
dmarc=pass (p=QUARANTINE sp=QUARANTINE dis=NONE) header.from=lovable.dev
DKIM-Signature: a=rsa-sha256; v=1; c=relaxed/relaxed; d=clou0109333.lovable.dev; q=dns/txt; s=mta; t=1756617778; x=1756624978; h=Message-Id: List-Unsubscribe-Post: List-Unsubscribe: To: To: From: From: Subject:
Subject: Content-Type: Mime-Version: Date: X-Feedback-Id: Sender: bh=P5z3d038F5K16hLPj5dtr/NF1u0R2ZE3hdU5PQ2anU+;
b=MDG7LRk6y11vcv170XUq36Jsh20u1210730455UQ/DWtYQV2F4d0R0V4W0V9f7eGv4d8d0e00V0H2CJH43V12K2CgVfSpwV1OxVfVj0VvAtb7mK5DcYALcd0I05CpQinJW0g9QVahTz=
DKIM-Signature: a=rsa-sha256; v=1; c=relaxed/relaxed; d=u.mailgun.org; q=dns/txt; s=kr2; t=1756617778; x=1756624978; h=Message-Id: List-Unsubscribe-Post: List-Unsubscribe: To: To: From: From: Subject: Subject:
Content-Type: Mime-Version: Date: X-Feedback-Id: Sender: bh=P5z3d038F5K16hLPj5dtr/NF1u0R2ZE3hdU5PQ2anU+;
b=FB0J40T0R1LJd0U320e0V0V046471gY2ZefwA7GLg/KP8B44konUQV718DQCyv4nqk3QmRV10Q0wV/LJgVtAdU3brkuTHWQ2251XJ3Jme3ot3eV1ZT1Gy2TRWahm407R032kx+F1/pclpChI0XFPACu2Q=
X-Mailgun-Sid: wY390w0CC1I0N1I0N1b4d4E1t0e0m04f0C5Jp004LCN2U3H0d3
X-Feedback-Id: postmaster@clou0109333.lovable.dev: 66978e735b18f899e3da5037:mailgun
Received: by 3c065b402945a1127d77a460d4c266382dc355d0273d81b3543c9abea89d8 with HTTP id 68b3dc32dcfa73ee68d3c; Sun, 31 Aug 2025 05:22:58 GMT
X-Mailgun-Sending-Ip: 143.55.239.140
Date: Sun, 31 Aug 2025 05:22:58 +0000
Mime-Version: 1.0
Content-Type: multipart/alternative; boundary="3097247993b1cfc1b21c61ed16eeeb5f54bee1356230b6a3f04fc54"
Subject: Some inspiration of what people are building with Lovable
From: Lovable <noreply@lovable.dev>
To: sudhama07pm@gmail.com
X-Mailgun-DKim: true
X-Mailgun-Native-Send: true
```

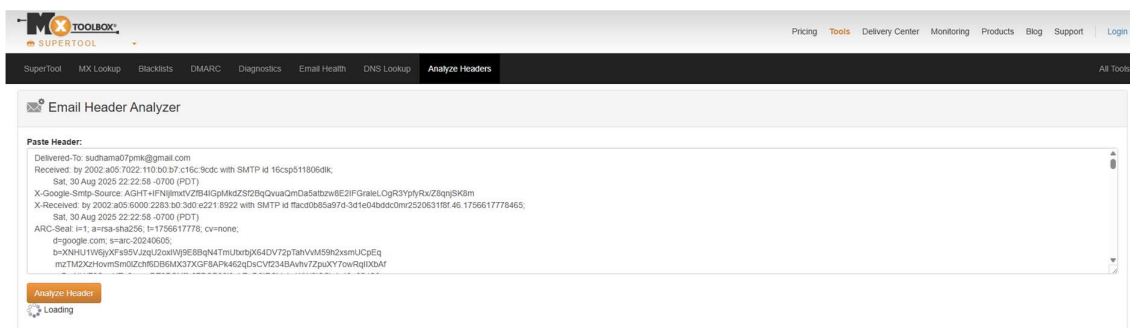
Step 2: Use a Mail Header Analyzer (MHA)

- The easiest way to analyze the header is with an online tool.
- Navigate to a site like MXToolbox Email Header Analyzer.



- Paste the entire header you copied into the analysis box.

Click the "Analyze" button to get a parsed, easy-to-read report.



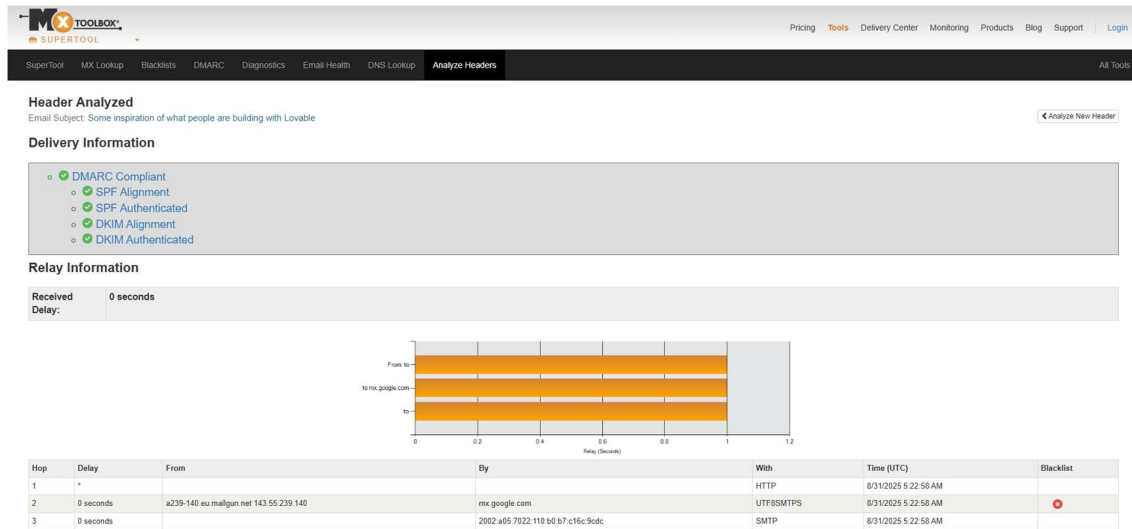
Step 3: Analyze The Report

- This is the most critical step. In the analyzer's report, look for the SPF, DKIM, and DMARC results.

Review the Overall Delivery Summary

- First, look at the high-level summary to get an immediate sense of the email's status.
- Check DMARC Compliance: The report shows the email is DMARC Compliant. This is the most important overall result.
- Check SPF Status: Both SPF Authenticated and SPF Alignment passed. This means the sending server was authorized.

- Check DKIM Status: DKIM Alignment passed, but DKIM Authenticated failed (✗). This is a critical finding and indicates a problem with the email's digital signature.



Investigate the SPF Record and Email Path

- Next, verify why the SPF check passed.
- Examine the SPF Record: The SPF record for the domain is v=spf1 include:mailgun.org ~all. This record explicitly authorizes servers from Mailgun to send emails on its behalf.
- Trace the Relay Information: The delivery path shows the email was sent from m241-136.mailgun.net.
- Conclusion: Since the email came from a Mailgun server, which is authorized in the SPF record, the SPF check passed.

SPF and DKIM Information

dmARC:lovable.dev
Show
Solve Email Delivery Problems

v=DMARC1; p=quarantine

spf:cloeu109333.lovable.dev:143.55.239.140
Show
Solve Email Delivery Problems

v=spf1 include:mailgun.org ~all

dkim:cloeu109333.lovable.dev:mta
Show

DKIM Policy Header

k=rsa; p=H0P9W6C5q051o30Q8a4A8Q0C3iQ8gQzCh18yYhXK1/187y0r5eW4RE3u4Ic4u3eM5d8d4kHw/2M3aJdE8C25L1P3j6fHucv45u2Kc/AD3yv9W6LBeB8twpY9PaQZ3kU8ly5AuyvvdqF1c2a2v4eVhrE3wqW4H9y1Y2a20P8wuhg9Z/6uN6R1VBoxIDaQ4

DKIM Signature

a=rsa-sha256; v=1; c=relaxed/relaxed; d=cloeu109333.lovable.dev; q=dns/txt; s=mta; t=1756627778; x=1756624978; h=Message-Id: List-Unsubscribe-Post: List-Unsubscribe: To: To: From: From: Subject: Subject: Content-Type: MIME-Version: Date: X-Feedback-Id: Sender; bh=Ps2h038FSK

dkim:eu.mailgun.org:krs2
Show

DKIM Policy Header

k=rsa; p=H0P9W6C5q051o30Q8a4A8Q0C3iQ8gQzCh18yYhXK1/187y0r5eW4RE3u4Ic4u3eM5d8d4kHw/2M3aJdE8C25L1P3j6fHucv45u2Kc/AD3yv9W6LBeB8twpY9PaQZ3kU8ly5AuyvvdqF1c2a2v4eVhrE3wqW4H9y1Y2a20P8wuhg9Z/6uN6R1VBoxIDaQ4

DKIM Signature

a=rsa-sha256; v=1; c=relaxed/relaxed; d=eu.mailgun.org; q=dns/txt; s=krs2; t=1756627778; x=1756624978; h=Message-Id: List-Unsubscribe-Post: List-Unsubscribe: To: To: From: From: Subject: Subject: Content-Type: MIME-Version: Date: X-Feedback-Id: Sender; bh=Ps2h038FSK10LP5d

Analyze the DKIM Failure

SPF and DKIM Information

dmarc:lovable.dev

Hide

Solve Email Delivery Problems

v=DMARC1; p=quarantine

Tag	TagValue	Name	Description
v	DMARC1	Version	Identifies the record retrieved as a DMARC record. It must be the first tag in the list.
p	quarantine	Policy	Policy to apply to email that fails the DMARC test. Valid values can be 'none', 'quarantine', or 'reject'.

Test	Result
✓ DMARC Record Published	DMARC Record found
✓ DMARC Syntax Check	The record is valid
✓ DMARC Multiple Records	Multiple DMARC records corrected to a single record.
✓ DMARC Policy Not Enabled	DMARC Quarantine/Reject policy enabled
✓ DMARC External Validation	All external domains in your DMARC record are giving permission to send them DMARC reports.

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Reported by [ivan.ns.cloudflare.com](#) on 9/1/2025 at 3:54:34 PM (UTC 0). [Just for you.](#)

Transcript